

Stage 6 — What to do instead of masking, and the state of the thesis

Where the Shortcut Sits: progress report 6

In one sentence. What to use instead of masking depends on what carries the shortcut: an annotation-free mask-then-erase method (U-MtE) repairs artifacts that carry the shortcut in their own pixels (calipers: thyroid Trap A +0.333 [+0.302, +0.364] versus ERM, masking +0.109 [+0.085, +0.133]), needs a disease-protection step when the artifact resembles the pathology (capsule debris), and cannot help when the shortcut is carried by correlates of the artifact (hair, chest drains), where only label-based reweighting works; with image-level artifact labels, U-MtE with balancing or masking with DFR is the most robust choice; for fully fine-tuned networks erasure must happen during training; and no remedy improves on masking uniformly on unaltered data. The thesis manuscript, its number audit, the CLAIM checklist and a blinded clinician audit package are ready; the audit itself has not been performed.

Notation

$Y \in \{0, 1\}$	diagnosis: 1 = melanoma, malignant nodule, tumour with a solid component, erosion, pneumothorax.
$A \in \{0, 1\}$	artifact present: a real artifact (hair, caliper, debris, drain) or a synthetic one in the controlled sweeps.
ROI	region of interest: lesion, nodule, tumour, union of expert lesion boxes, lungs; drawn in green; artifacts in red.
r	overlap $ \text{artifact} \cap \text{ROI} / \text{artifact} $. Trap A: $r \geq 0.5$ (artifact in the ROI); Trap B: $r < 0.1$ (outside).
Environments	training and correlated test $P(A=1 Y=1) = 0.9$, $P(A=1 Y=0) = 0.1$; reversed test 0.1/0.9; clean test: artifact independent of Y (real artifacts: half of each class) or absent (synthetic artifacts).
ERM, mask	linear head on the frozen features of the original image / of the ROI-masked image.
Crossover	$[\text{mask} - \text{ERM}]_{\text{outside}} - [\text{mask} - \text{ERM}]_{\text{inside}}$ of reversed-test AUROC.
CI	95% crossed seed×image bootstrap interval (Stage 1); analyses whose per-image predictions were not saved are shown as point estimates and marked.
$ \Delta p $	same-head counterfactual sensitivity: change of the predicted probability when the artifact is inserted into the same image.
U-MtE	mask-then-erase: ROI masking followed by erasure of the feature subspace along which <i>generic</i> synthetic overlays move masked images (Sec. 3).
$\min(\text{rev}, \text{corr})$	robustness summary: the lower of reversed and correlated AUROC; an arm that flips the shortcut (correlated < reversed -0.02) is marked [†] .

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1 Where this stage starts

Stages 1–5 established that ROI masking removes an artifact outside the ROI and concentrates the model on one inside it; that this is caused by location, not by which images carry the artifact; that it matters on unaltered data and at clinical operating points; and that a two-parameter model explains it. They also showed three facts that shape any remedy: label-based methods (balancing, DFR) fix both locations but need image-level artifact labels; removing the artifact’s pixels fails when the artifact resembles the disease (capsule debris); and some shortcuts are carried by correlates of the artifact rather than its pixels (hair, drains). This stage asks what a practitioner should do instead of masking, and reports the state of the thesis.

2 Questions

- Q6a** Can an in-ROI shortcut be removed from frozen foundation-model features *without* any artifact annotation — no artifact example, mask or label, only the ROI mask that masking already uses?
- Q6b** When does such a remedy fail, and which alternative fits which regime?
- Q6c** Can the same be done for networks fine-tuned end to end?
- Q6d** Does any remedy transfer to unaltered test data?
- Q6e** How will the automatic artifact labels be validated by a clinician, and what remains before submission?

3 Method: from insert-then-erase to U-MtE

Idea Masking leaves exactly one thing to remove: what lies inside the ROI and is not disease. If we knew the directions in feature space along which an artifact moves an image, we could project them out. The method estimates those directions by *inserting* artifacts into images and measuring how the features change.

Evolution

1. **Insert-then-erase (I2E).** Paste artifact *templates* (e.g. real hair strands cut from other images) into training images, collect the paired feature differences (with minus without), take the subspace U_k holding 90% of their held-out energy (cap 64 directions), and erase it: $x \mapsto x - U_k U_k^\top (x - \mu)$; a logistic head is trained on the erased features. Works, but needs examples of the artifact.
2. **Mask-then-erase (MtE).** Estimate the subspace on *masked* images (insert, then mask), so that erasure targets exactly what masking leaves; the head is trained and tested on masked, erased features.
3. **U-MtE (universal).** Replace the templates by one *generic* procedural overlay library — thin strokes, thick bands, solid and translucent patches, blobs, dashed lines, small glyphs; random colour (half grey), opacity, width and position, half centred in the ROI (Figure 1) — used *unchanged* for every artifact, dataset and modality. It contains no ruler, no caliper and no hair. U-MtE needs no artifact example, mask or label; only the ROI mask.
4. **Disease protection.** If the erased subspace overlaps the disease direction, erasure removes disease evidence. Protection estimates label directions W on the *artifact-free* training images and erases $\text{orth}((I - WW^\top)U_k)$, which leaves every protected direction untouched ($w^\top P(x) = w^\top x$). It needs diagnosis labels of artifact-free images (which any training set has) and no artifact label.
5. **U-MtE with balancing.** When image-level artifact labels exist, the head is group-balanced.

Why it should work (theory of Stage 5) Erasure sets the effective artifact signal $\tilde{A}$ to zero but scales the disease signal by $1 - \rho^2$, where ρ is the overlap between the erased subspace and the disease direction. It therefore beats in-ROI masking unless ρ is large — which is the case when the artifact resembles the disease, and which protection forces to zero along the protected directions. Overlay *augmentation* with the same overlays only decorrelates the label from the generated overlays, not from the real artifact; erasure removes the whole subspace of “foreign overlay” changes, which contains the real artifact’s direction when both move the representation in shared ways. The theory therefore predicts erase $\gg$ augment for overlay-like artifacts.

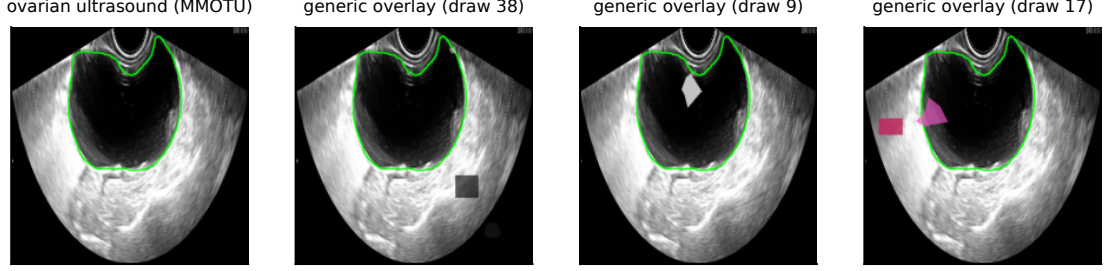


Figure 1: The generic overlay library (the same for every modality) drawn on an ovarian ultrasound image (MMOTU, CC BY 4.0; green: ROI). Overlays are inserted, then the image is masked, to estimate the subspace that U-MtE erases.

Baselines JTT (Liu et al., 2021) (label-free: up-weight the training errors of an ERM model); SPLINCE (Holstege et al., 2025) (covariance-preserving linear concept removal with artifact labels; faithful whitened implementation, verified against the paper’s theorem); LEACE (Stages 1 and 3); LaMa inpainting (Suvorov et al., 2022) of the artifact pixels (needs artifact masks); text-prompted erasure with vision-language models (the artifact direction from a text description, Chuang et al., 2023); SLAS, a few-shot patch-token artifact localiser (5–50 annotated images); pseudo-group balancing with groups inferred by an overlay detector; erasure followed by DFR; and two validation-driven selectors that pick, per fold, the candidate with the best worst-group validation score.

Fine-tuned networks `mte_ft`: masked images with and without a generic overlay, plus a penalty on the distance of their normalised penultimate features; `umte_ft` (erase-while-fine-tuning): a running overlay-shift subspace is estimated on the network’s own features and projected out before the head at every step, plus the invariance penalty; `cons_ft`: prediction consistency with and without an overlay; `umte_cons_ft`: all three; `mte_post`: fine-tune like masking, then apply U-MtE to the penultimate features.

Tool Both methods are packaged for any backbone: `python -m wtss.umte fit/embed` (images and ROI masks only) and `python -m wtss.slas fit/embed` (a few annotated images).

4 Experiments

Table 1: Experiments of this stage and their registration documents (docs/PREREGISTRATION_<name>.md or docs/<name>.md). Commit times are set by the committing machine and are not independent proof of order.

Experiment	Status	Registration	First commit (local time)	Support criterion
U-I2E / U-MtE, protection, balancing, U7–U14	supported registered (with amendments before each run)	UNIVERSAL_I2E	ee95f10, 20:14	2026-09-26 per claim, CI; no gaming (clean loss ≤ 0.02 , corr $\geq$ rev -0.02)
Remedy tests P2–P4 in the 16-test family	partly supported tests registered; family grouped post hoc	STATISTICAL_PLAN	4e9d44c, 09:37	2026-09-27 Holm $p < 0.05$, estimate > 0
Erasure-rank ablation	supported registered	UMTE_ABLATION	30f19c8, 14:15	2026-09-27 descriptive
Encoder size (S2)	supported registered	SCALE	13303b0, 14:13	2026-09-27 protected U-MtE $>$ mask, CI
LaMa inpainting	supported registered	INPAINT_LAMA	715b790, 09:41	2026-09-27 reported with CI, no direction
Text-prompted erasure (V1–V3)	supported registered	TEXT_PROMPT	7877e8c, 12:21	2026-09-27 V1: text erasure $>$ mask
SLAS (S0–S6)	partly supported registration and localisation result share a commit	SLAS	2d1ba20, 23:33	2026-09-26 S1: mask-then-suppress $>$ token masking
Chest drains (D1–D5)	supported registered	CXR_DRAIN	3d0cffc, 03:29	2026-09-27 U-MtE $>$ mask, CI
Fine-tuning F1–F3	supported registered	FINETUNE	7c8b466, 07:06	2026-09-27 mte_ft $>$ mask, CI
Fine-tune, then erase	supported registered	FT_ERASE	8ac7a7d, 14:20	2026-09-27 mte_post $>$ mask_post
Erase while fine-tuning (G1–G4)	supported registered (amendments before running)	FT_UMTE	b7d7077, 14:52	2026-09-27 umte(_cons)_ft $>$ mask for both architectures
Leakage: embedding groups (P2–P4)	supported registered	EMBEDDING_GROUPS	b0eda23, 13:55	2026-09-27 same sign and CI verdict
Natural data (N1, N2, N4)	supported registered	NATURAL	74bd243, 09:39	2026-09-27 protected U-MtE $\geq$ mask
Clinician image audit (prepared)	supported registered	FINAL (Part B)	c45d482, 00:35	2026-09-28 precision, recall, location agreement, κ

Every remedy analysis was registered before it was run (docs/PREREGISTRATION_UNIVERSAL_I2E.md and its amendments U7–U14; fine-tuning in PREREGISTRATION_FT_ERASE.md and PREREGISTRATION_FT_UMTE.md; the rank ablation, LaMa, text prompts, SLAS and the chest drains in their own documents). Every outcome is reported whichever way it went.

5 Results

5.1 Remedies in the real traps

Figure 2 and Table 2 (DINOv2, Trap A); Table 3 gives every arm, both traps and three encoder families.

- **Calipers (pixel-carried).** Without any annotation, U-MtE repairs most of what masking leaves in thyroid (+0.333 [+0.302, +0.364] versus ERM; masking +0.109 [+0.085, +0.133]), with MedSigLIP (+0.299 [+0.270, +0.328]) and with ConvNeXt (+0.131 [+0.087, +0.171]), and in the

ovary with MedSigLIP (+0.212 [+0.163, +0.260]). With DINOv2 in the ovary its gain is small (+0.152 [+0.098, +0.203]; protected +0.175 [+0.116, +0.235]).

- **Debris (pathology-like).** Unprotected U-MtE *lowers* capsule performance (−0.042 [−0.076, −0.009]; MedSigLIP −0.113 [−0.146, −0.080]): debris resembles erosion fibrin, so the overlay subspace overlaps the disease. Protection repairs this (+0.078 [+0.047, +0.107]) but does not beat masking; label-based methods are best.
- **Hair (correlate-carried).** No pixel-level method helps much (U-MtE −0.009 [−0.043, +0.024] versus ERM, i.e. +0.040 [+0.024, +0.056] over masking); balancing (+0.213 [+0.183, +0.243]) and DFR (+0.196 [+0.157, +0.235]) are required.
- **Outside the ROI (Trap B)** U-MtE neither helps nor harms relative to masking — masking has already removed the artifact.

Table 2: Remedies in the in-ROI trap: change in reversed-test AUROC versus ERM (DINOv2, 5 seeds $\times$ 5 folds, crossed CIs). U-MtE needs no artifact annotation; balancing and DFR need image-level artifact labels.

Cohort (Trap A, in-ROI)	ROI masking	U-MtE	U-MtE protected	group balancing	DFR	U-MtE + balancing
Dermoscopy hair	−0.049 [−0.083, −0.018]	−0.009 [−0.043, +0.024]	−0.008 [−0.046, +0.028]	+0.213 [+0.183, +0.243]	+0.196 [+0.157, +0.235]	+0.185 [+0.145, +0.224]
Thyroid calipers	+0.109 [+0.085, +0.133]	+0.333 [+0.302, +0.364]	+0.393 [+0.347, +0.434]	+0.388 [+0.360, +0.418]	+0.463 [+0.436, +0.492]	+0.498 [+0.471, +0.524]
Ovarian calipers	+0.132 [+0.072, +0.196]	+0.152 [+0.098, +0.203]	+0.175 [+0.116, +0.235]	+0.290 [+0.248, +0.332]	+0.310 [+0.263, +0.357]	+0.321 [+0.271, +0.372]
Capsule debris	+0.084 [+0.055, +0.112]	−0.042 [−0.076, −0.009]	+0.078 [+0.047, +0.107]	+0.179 [+0.150, +0.205]	+0.234 [+0.199, +0.273]	+0.258 [+0.221, +0.293]

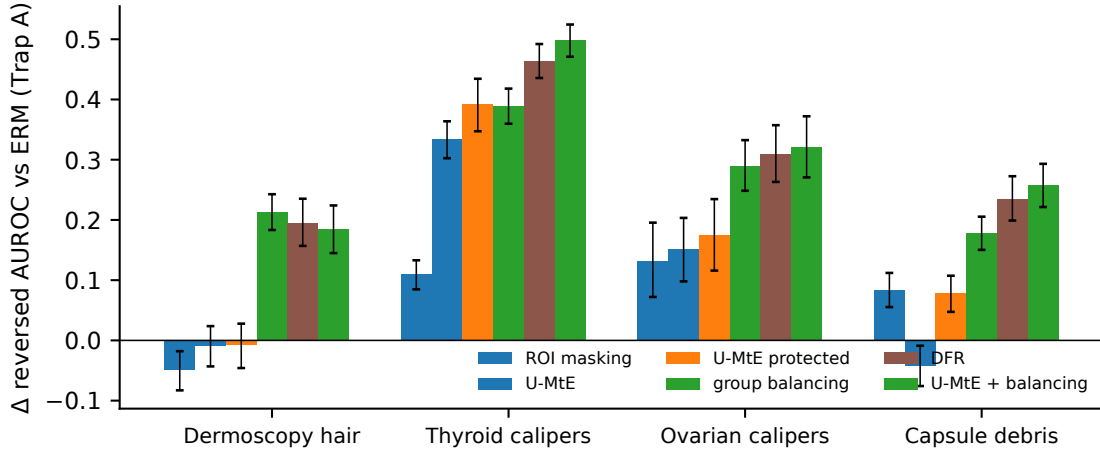


Figure 2: Remedies in the in-ROI trap (change in reversed AUROC versus ERM, DINOv2, crossed 95% CIs).

Table 3: Every remedy in every real-artifact trap and encoder: change in reversed-test AUROC versus ERM (5 seeds $\times$ 5 grouped folds, crossed CIs). U-MtE and the augmentation arm use only the ROI mask; balancing, DFR and their combinations need image-level artifact labels.

Cohort, encoder	Arm — ERM	Trap A (in ROI)	Trap B (outside)
Dermoscopy hair, DINOv2	masking	−0.049 [−0.083, −0.018]	+0.098 [+0.060, +0.136]
	U-MtE	−0.009 [−0.043, +0.024]	+0.121 [+0.083, +0.160]
	U-MtE protected	−0.008 [−0.046, +0.028]	+0.124 [+0.077, +0.170]
	same overlays as augmentation	−0.044 [−0.078, −0.012]	+0.107 [+0.070, +0.145]
	group balancing	+0.213 [+0.183, +0.243]	+0.204 [+0.166, +0.245]
	DFR	+0.196 [+0.157, +0.235]	+0.196 [+0.147, +0.251]
	masking + DFR	+0.152 [+0.103, +0.202]	+0.206 [+0.142, +0.267]
Thyroid calipers, DINOv2	U-MtE + balancing	+0.185 [+0.145, +0.224]	+0.240 [+0.197, +0.283]
	masking	+0.109 [+0.085, +0.133]	+0.348 [+0.307, +0.388]
	U-MtE	+0.333 [+0.302, +0.364]	+0.347 [+0.308, +0.386]

Cohort, encoder			Arm – ERM	Trap A (in ROI)	Trap B (outside)
Thyroid calipers, Med-SigLIP			U-MtE protected	+0.393 [+0.347, +0.434]	+0.349 [+0.310, +0.389]
			same overlays as augmentation	+0.118 [+0.094, +0.143]	+0.345 [+0.305, +0.385]
			group balancing	+0.388 [+0.360, +0.418]	+0.254 [+0.218, +0.292]
			DFR	+0.463 [+0.436, +0.492]	+0.312 [+0.262, +0.362]
			masking + DFR	+0.537 [+0.506, +0.567]	+0.357 [+0.313, +0.400]
			U-MtE + balancing	+0.498 [+0.471, +0.524]	+0.371 [+0.329, +0.412]
			masking	+0.053 [+0.032, +0.073]	+0.248 [+0.200, +0.293]
			U-MtE	+0.299 [+0.270, +0.328]	+0.276 [+0.232, +0.318]
			U-MtE protected	+0.358 [+0.330, +0.388]	+0.271 [+0.224, +0.318]
			same overlays as augmentation	+0.074 [+0.053, +0.095]	+0.248 [+0.202, +0.293]
Thyroid calipers, Con-vNeXt			group balancing	+0.409 [+0.382, +0.436]	+0.201 [+0.161, +0.241]
			DFR	+0.463 [+0.437, +0.489]	+0.243 [+0.207, +0.282]
			masking + DFR	+0.517 [+0.490, +0.545]	+0.279 [+0.234, +0.324]
			U-MtE + balancing	+0.505 [+0.478, +0.531]	+0.313 [+0.269, +0.358]
			masking	−0.004 [−0.029, +0.020]	+0.271 [+0.228, +0.314]
			U-MtE	+0.131 [+0.087, +0.171]	+0.276 [+0.231, +0.320]
			U-MtE protected	+0.179 [+0.112, +0.236]	+0.286 [+0.241, +0.332]
			same overlays as augmentation	+0.003 [−0.021, +0.027]	+0.270 [+0.227, +0.313]
			group balancing	+0.326 [+0.305, +0.348]	+0.222 [+0.186, +0.258]
			DFR	+0.369 [+0.335, +0.405]	+0.260 [+0.212, +0.305]
Ovarian calipers, DINOv2			masking + DFR	+0.408 [+0.372, +0.442]	+0.307 [+0.259, +0.354]
			U-MtE + balancing	+0.397 [+0.362, +0.434]	+0.313 [+0.266, +0.358]
			masking	+0.132 [+0.072, +0.196]	+0.302 [+0.225, +0.378]
			U-MtE	+0.152 [+0.098, +0.203]	+0.317 [+0.242, +0.388]
			U-MtE protected	+0.175 [+0.116, +0.235]	+0.322 [+0.243, +0.398]
			same overlays as augmentation	+0.117 [+0.054, +0.178]	+0.305 [+0.224, +0.383]
			group balancing	+0.290 [+0.248, +0.332]	+0.256 [+0.190, +0.320]
			DFR	+0.310 [+0.263, +0.357]	+0.278 [+0.209, +0.349]
			masking + DFR	+0.339 [+0.284, +0.393]	+0.347 [+0.264, +0.428]
			U-MtE + balancing	+0.321 [+0.271, +0.372]	+0.343 [+0.265, +0.419]
Ovarian calipers, Med-SigLIP			masking	+0.002 [−0.048, +0.051]	+0.180 [+0.120, +0.238]
			U-MtE	+0.212 [+0.163, +0.260]	+0.191 [+0.136, +0.245]
			U-MtE protected	+0.155 [+0.098, +0.209]	+0.166 [+0.108, +0.224]
			same overlays as augmentation	+0.018 [−0.027, +0.062]	+0.179 [+0.123, +0.235]
			group balancing	+0.302 [+0.254, +0.351]	+0.149 [+0.103, +0.197]
			DFR	+0.370 [+0.320, +0.420]	+0.210 [+0.154, +0.265]
			masking + DFR	+0.356 [+0.299, +0.411]	+0.229 [+0.161, +0.296]
			U-MtE + balancing	+0.364 [+0.312, +0.416]	+0.219 [+0.161, +0.277]
			masking	+0.084 [+0.055, +0.112]	+0.461 [+0.415, +0.506]
			U-MtE	−0.042 [−0.076, −0.009]	+0.441 [+0.396, +0.483]
Capsule debris, DINOv2			U-MtE protected	+0.078 [+0.047, +0.107]	+0.442 [+0.397, +0.486]
			same overlays as augmentation	+0.096 [+0.067, +0.126]	+0.462 [+0.417, +0.505]
			group balancing	+0.179 [+0.150, +0.205]	+0.244 [+0.216, +0.272]
			DFR	+0.234 [+0.199, +0.273]	+0.287 [+0.239, +0.335]
			masking + DFR	+0.312 [+0.275, +0.349]	+0.446 [+0.393, +0.497]
			U-MtE + balancing	+0.258 [+0.221, +0.293]	+0.471 [+0.428, +0.514]
			masking	+0.117 [+0.088, +0.147]	+0.452 [+0.407, +0.495]
			U-MtE	−0.113 [−0.146, −0.080]	+0.406 [+0.362, +0.450]
			U-MtE protected	+0.088 [+0.055, +0.121]	+0.448 [+0.404, +0.491]
			same overlays as augmentation	+0.136 [+0.106, +0.166]	+0.451 [+0.407, +0.495]
Capsule debris, Med-SigLIP			group balancing	+0.225 [+0.203, +0.248]	+0.240 [+0.211, +0.269]
			DFR	+0.254 [+0.222, +0.286]	+0.279 [+0.242, +0.320]
			masking + DFR	+0.345 [+0.312, +0.378]	+0.438 [+0.391, +0.483]
			U-MtE + balancing	+0.269 [+0.232, +0.308]	+0.453 [+0.410, +0.496]
			masking	+0.084 [+0.049, +0.118]	+0.403 [+0.358, +0.448]
			U-MtE	+0.009 [−0.024, +0.040]	+0.378 [+0.334, +0.422]
			U-MtE protected	+0.108 [+0.074, +0.142]	+0.383 [+0.337, +0.429]
			same overlays as augmentation	+0.099 [+0.066, +0.131]	+0.406 [+0.361, +0.449]
			group balancing	+0.197 [+0.172, +0.221]	+0.226 [+0.198, +0.256]
			DFR	+0.255 [+0.223, +0.291]	+0.293 [+0.234, +0.346]
Capsule debris, Con-vNeXt			masking + DFR	+0.330 [+0.296, +0.364]	+0.453 [+0.403, +0.502]
			U-MtE + balancing	+0.273 [+0.231, +0.312]	+0.433 [+0.384, +0.480]

5.2 Which model is most robust?

A reversed-test gain can come from inverting the shortcut rather than removing it. Table 4 therefore reports min(reversed, correlated) AUROC. With image-level artifact labels, U-MtE with balancing is the most robust model for thyroid with all three encoders (e.g. 0.787 with DINOv2) and for the ovary with MedSigLIP (0.790); masking with DFR is best for capsule (0.892) and the DINOv2 ovary (0.752); plain balancing for hair (0.722) and DFR for chest drains (0.666). DFR and masking + DFR over-correct on thyroid: their correlated AUROC falls below the reversed one (marked [†]). Without artifact labels, protected U-MtE is the most robust label-free arm for thyroid; for capsule the best label-free choice is U-MtE combined with JTT (Table 11).

Table 4: Robustness summary in the in-ROI trap: min(reversed, correlated) AUROC per arm (mean over seeds; best per row in bold; [†] = the arm flips the shortcut, correlated < reversed −0.02).

Trap A	ERM	mask	U-MtE	U-MtE prot.	overlay aug.	JTT	balanced	DFR	mask+DFR	U-MtE+bal.
Dermoscopy hair, DINOv2	0.510	0.460	0.500	0.502	0.466	0.575	0.722	0.705	0.662	0.695
Thyroid calipers, DINOv2	0.289	0.398	0.622	0.682	0.407	0.417	0.677	0.669 [†]	0.751 [†]	0.787
Thyroid calipers, MedSigLIP	0.289	0.341	0.587	0.647	0.362	—	0.698	0.722 [†]	0.772 [†]	0.793
Thyroid calipers, ConvNeXt	0.357	0.353	0.488	0.536	0.361	—	0.684	0.669 [†]	0.713 [†]	0.755
Ovarian calipers, DINOv2	0.413	0.545	0.565	0.588	0.530	0.440	0.703	0.689 [†]	0.752	0.734
Ovarian calipers, MedSigLIP	0.426	0.428	0.638	0.581	0.444	—	0.729	0.756 [†]	0.782	0.790
Capsule debris, DINOv2	0.581	0.665	0.539	0.659	0.678	0.596	0.760	0.816	0.892	0.839
Capsule debris, MedSigLIP	0.570	0.687	0.457	0.658	0.706	—	0.796	0.825	0.915	0.839
Capsule debris, ConvNeXt	0.528	0.612	0.537	0.636	0.627	—	0.725	0.783	0.858	0.801
Chest drains, RAD-DINO	0.194	0.232	0.272	0.275	0.236	0.179	0.540	0.666	0.635	0.461

5.3 The 16-test primary family

Table 5. The four location tests (P1) and twelve remedy tests — protected U-MtE > mask (P2), erase > augment with the same overlays (P3), U-MtE+balancing > balancing (P4), in each of the four cohorts — were grouped into one Holm family after the individual results were known (conservative bookkeeping, not a pre-registered family). 11 of 16 are supported: all four P1; P2 in hair, thyroid and ovary (not capsule); P3 in hair and thyroid (capsule significant in the opposite direction; ovary not after Holm); P4 in thyroid and capsule.

Table 5: The 16-test primary family (DINOv2, Trap A reversed test unless stated): the four location tests and the twelve remedy tests, Holm-corrected together.

Family	Cohort	Estimate [95% CI]	<i>p</i>	<i>p</i> (Holm, 16)	Supported
P1 location crossover	ISIC hair (DINOv2)	+0.147 [+0.111, +0.185]	0.0001	0.0016	supported
P1 location crossover	Thyroid (DINOv2)	+0.239 [+0.196, +0.280]	0.0001	0.0016	supported
P1 location crossover	Capsule (DINOv2)	+0.377 [+0.327, +0.426]	0.0001	0.0016	supported
P1 location crossover	Ovary (DINOv2)	+0.170 [+0.087, +0.252]	0.0001	0.0016	supported
P2 U-MtE (protected) $\hat{z}$ mask	ISIC hair	+0.042 [+0.024, +0.058]	0.0001	0.0016	supported
P2 U-MtE (protected) $\hat{z}$ mask	Thyroid	+0.283 [+0.237, +0.329]	0.0001	0.0016	supported
P2 U-MtE (protected) $\hat{z}$ mask	Capsule	−0.006 [−0.018, +0.006]	0.2946	0.3486	not supported
P2 U-MtE (protected) $\hat{z}$ mask	Ovary	+0.043 [+0.023, +0.065]	0.0001	0.0016	supported
P3 erase $\hat{z}$ augment	ISIC hair	+0.035 [+0.017, +0.055]	0.0002	0.0016	supported
P3 erase $\hat{z}$ augment	Thyroid	+0.215 [+0.189, +0.244]	0.0001	0.0016	supported
P3 erase $\hat{z}$ augment	Capsule	−0.139 [−0.158, −0.120]	0.0001	0.0016	not supported (opposite)
P3 erase $\hat{z}$ augment	Ovary	+0.035 [+0.003, +0.067]	0.0326	0.1304	not supported
P4 U-MtE+balanced $\hat{z}$ balanced	ISIC hair	−0.028 [−0.064, +0.007]	0.1162	0.3486	not supported
P4 U-MtE+balanced $\hat{z}$ balanced	Thyroid	+0.110 [+0.079, +0.142]	0.0001	0.0016	supported
P4 U-MtE+balanced $\hat{z}$ balanced	Capsule	+0.079 [+0.042, +0.115]	0.0002	0.0016	supported
P4 U-MtE+balanced $\hat{z}$ balanced	Ovary	+0.031 [−0.014, +0.077]	0.1730	0.3486	not supported

5.4 Remedies in the controlled sweeps

Tables 6 and 7. With DINOv2 and RAD-DINO, U-MtE beats masking at full overlap in every modality sweep (thyroid +0.278 [+0.214, +0.344], capsule +0.121 [+0.078, +0.170], chest tube +0.134 [+0.113, +0.156]). With MedSigLIP it fails on both the synthetic caliper and the synthetic debris; protection turns the debris failure into a gain (+0.114 [+0.022, +0.238]) but not the caliper

failure ($-0.041 [-0.095, +0.002]$). U-MtE with balancing beats masking in every sweep and has no positive location interaction (dermoscopy: $-0.083 [-0.114, -0.050]$, against masking’s large positive one). On the dermoscopy ruler, the insert-then-erase precursor on unmasked features helps ($+0.197 [+0.146, +0.247]$) while plain U-MtE does not ($-0.007 [-0.085, +0.070]$): with the synthetic ruler, the subspace estimated from generic overlays evidently misses the direction of the retained ruler, and only the balanced variant repairs it.

Table 6: Remedies in the controlled sweeps at full overlap (reversed-test AUROC, crossed CIs).

Controlled sweep, $r = 1$	U-MtE – mask	U-MtE protected – mask	U-MtE+bal. – mask	U-MtE+bal. – balanced
Thyroid caliper, DINOv2	+0.278 [+0.214, +0.344]	+0.287 [+0.141, +0.464]	+0.470 [+0.391, +0.550]	+0.091 [+0.013, +0.172]
Thyroid caliper, MedSigLIP	−0.076 [−0.120, −0.034]	−0.041 [−0.095, +0.002]	+0.458 [+0.393, +0.525]	+0.069 [+0.002, +0.136]
Ovarian caliper, DINOv2	+0.140 [+0.061, +0.225]	+0.114 [+0.043, +0.192]	+0.251 [+0.157, +0.355]	−0.032 [−0.165, +0.104]
Capsule debris, DINOv2	+0.121 [+0.078, +0.170]	+0.130 [+0.083, +0.189]	+0.211 [+0.134, +0.297]	+0.057 [−0.008, +0.121]
Capsule debris, MedSigLIP	−0.190 [−0.265, −0.120]	+0.114 [+0.022, +0.238]	+0.299 [+0.195, +0.398]	−0.018 [−0.101, +0.065]
Chest tube, RAD-DINO	+0.134 [+0.113, +0.156]	+0.144 [+0.125, +0.164]	+0.168 [+0.145, +0.191]	−0.031 [−0.042, −0.021]

Table 7: The dermoscopy ruler sweep at full overlap: the insert-then-erase precursor (U-I2E, erasing on unmasked features) and the mask-then-erase variants, versus ERM (crossed CIs).

Dermoscopy ruler sweep, $r = 1$ (DINOv2)	U-I2E – ERM	U-MtE – ERM	U-MtE protected – ERM	U-MtE+bal. – ERM
change in reversed AUROC	+0.197 [+0.146, +0.247]	−0.007 [−0.085, +0.070]	+0.091 [−0.016, +0.185]	+0.243 [+0.167, +0.317]

5.5 Erasure versus augmentation, protection, and other contrasts

Table 8. The *same* overlays used as training augmentation instead of for erasure gain almost nothing over masking (Table 3, “same overlays as augmentation”), while erasure gains $+0.225 [+0.202, +0.247]$ over augmentation with MedSigLIP and $+0.128 [+0.093, +0.161]$ with ConvNeXt on thyroid: it is the subspace estimation, not exposure to overlays, that helps. Protection is a safety net: it costs nothing on hair ($+0.001 [-0.016, +0.017]$), helps thyroid ($+0.059 [+0.019, +0.096]$) and removes the capsule failure (protected – mask $-0.006 [-0.018, +0.006]$ versus unprotected $-0.126 [-0.146, -0.107]$).

Table 8: Secondary remedy contrasts (reversed test, crossed CIs).

Cohort	Contrast (Trap A)	Estimate [95% CI]
hair	mte – mask	+0.040 [+0.024, +0.056]
hair	mte_protect – mte	+0.001 [−0.016, +0.017]
thyroid	mte_protect – mte	+0.059 [+0.019, +0.096]
thyroid	mte – jtt	+0.206 [+0.171, +0.241]
ovary	mte – jtt	+0.125 [+0.063, +0.189]
thyroid MedSigLIP	mte – mte_aug	+0.225 [+0.202, +0.247]
thyroid ConvNeXt	mte – mte_aug	+0.128 [+0.093, +0.161]
thyroid ConvNeXt	mte – mask	+0.135 [+0.102, +0.168]
capsule	mte – mask	−0.126 [−0.146, −0.107]
capsule	mte_protect – mask	−0.006 [−0.018, +0.006]

5.6 How many directions must be erased?

Table 9. On thyroid the gain is robust for every setting that erases ≥ 16 directions or $\geq 50\%$ of the overlay-shift energy; one to four directions recover only a third of it — the overlay shift is multi-dimensional. In these cohorts the 90% rule reaches the cap of 64 directions. Unprotected erasure harms capsule more the more directions it removes, as the theory’s ρ predicts; the protected variant never harms capsule and helps the ovary at every rank.

Table 9: Erasure-rank ablation (DINOv2, Trap A): reversed-AUROC difference to masking (mean over seeds) for the energy rule (share of held-out overlay-shift energy removed; cap 64) and for fixed ranks k .

Erased rank	Thyroid U-MtE	Ovary U-MtE	Ovary protected	Capsule U-MtE	Capsule protected
energy 0.50	+0.190	−0.003	+0.033	−0.067	−0.006
energy 0.90 (main)	+0.224	+0.020	+0.043	−0.126	−0.006
energy 0.99	+0.224	+0.020	+0.043	−0.126	−0.006
$k = 1$	+0.068	+0.010	+0.009	+0.001	+0.000
$k = 4$	+0.056	−0.038	+0.015	−0.010	−0.003
$k = 16$	+0.183	−0.011	+0.020	−0.058	−0.009
$k = 64$	+0.224	+0.020	+0.043	−0.126	−0.006

5.7 Encoder size

Table 10. Protected U-MtE beats masking in thyroid and ovary at every DINOv2 size; on capsule protection is less complete with ViT-L.

Table 10: U-MtE across encoder sizes (crossed CIs).

Cohort	DINOv2	U-MtE − mask (Trap A)	U-MtE protected − mask
Thyroid	ViT-S/14	+0.192 [+0.159, +0.227]	+0.225 [+0.168, +0.276]
Thyroid	ViT-B/14	+0.224 [+0.195, +0.254]	+0.283 [+0.237, +0.329]
Thyroid	ViT-L/14	+0.246 [+0.216, +0.279]	+0.274 [+0.243, +0.306]
Ovary	ViT-S/14	+0.121 [+0.084, +0.158]	+0.103 [+0.074, +0.134]
Ovary	ViT-B/14	+0.020 [−0.016, +0.054]	+0.043 [+0.023, +0.065]
Ovary	ViT-L/14	+0.094 [+0.061, +0.127]	+0.060 [+0.037, +0.084]
Capsule	ViT-S/14	−0.153 [−0.172, −0.134]	−0.002 [−0.018, +0.016]
Capsule	ViT-B/14	−0.126 [−0.146, −0.107]	−0.006 [−0.018, +0.006]
Capsule	ViT-L/14	−0.097 [−0.114, −0.079]	−0.044 [−0.056, −0.032]

5.8 Published baselines and alternatives

Table 11 (crossed intervals), Table 12 (analyses first archived without per-image predictions and re-generated with them; crossed intervals) and Table 13 (point estimates of analyses whose predictions were not saved).

- **JTT** (label-free): U-MtE beats it on thyroid (+0.206 [+0.171, +0.241]) and ovary (+0.125 [+0.063, +0.189]); JTT wins on hair (min(rev, corr) 0.564 versus 0.506); on capsule U-MtE+JTT is the most robust label-free combination (0.787).
- **SPLINCE** (needs artifact labels): the faithful implementation removes much disease signal (clean AUROC 0.577 on thyroid) and still inverts the shortcut on capsule; U-MtE with balancing is more robust on thyroid, capsule and hair (0.787 versus 0.565; 0.839 versus 0.668; 0.695 versus 0.575).
- **LaMa inpainting** of the detected artifact pixels, followed by masking, is a strong baseline that needs artifact masks (over masking: thyroid +0.224 [+0.206, +0.242], ovary +0.128 [+0.084, +0.169]). Protected U-MtE, which needs no artifact mask, exceeds it on thyroid (+0.060 [+0.013, +0.105]) and is below it on ovary (−0.085 [−0.120, −0.050]) and hair (−0.086, archived point estimate); U-MtE with balancing beats it everywhere. The thyroid comparison changed when the run was regenerated (archived −0.009): the protected arm reproduced about 0.06 higher than in its first run, with unchanged code for that arm, and now agrees with the protected arm of the main regenerated run ([results/bootstrap-correction/archived_old_vs_new.md](#)).
- **Text-prompted erasure** with MedSigLIP and DermLIP is a near no-op ($\leq +0.008$ over masking): the text-space artifact direction does not coincide with the image-space direction along which the artifact moves the representation.

- **SLAS** localises hair from a few annotated images, but the gain it enables is capped by what hair pixels carry: removing *every* real hair patch (an oracle) gains only +0.015 over token-level masking.
- **Pseudo-group balancing** from an overlay detector helps on thyroid (reversed 0.663 versus 0.622 for U-MtE) and not on hair, where the detector cannot see hair.
- **Erasure + DFR** adds at most about 0.01 over masking + DFR: once a group-robust head is used, masking does the rest.
- **Validation-driven selectors** beat masking in every cohort and are within about 0.02 of the best candidate, but neither is uniformly best; admitting DFR as a candidate fixes chest drains and makes the choice noisier elsewhere.
- **Chest drains**: U-MtE gains +0.040 [+0.023, +0.057] over masking with RAD-DINO and nothing with DINOv2 (−0.001 [−0.009, +0.008]); DFR is best.
- **Templates instead of generic overlays** on capsule debris make erasure worse (−0.392 [−0.425, −0.360] against masking); disease protection removes that failure (+0.003 [−0.014, +0.021]).

Table 11: Published baselines: JTT (label-free error upweighting) and SPLINCE (faithful whitened implementation of the covariance-preserving concept projection, needs artifact labels), beside U-MtE (mean over seeds).

Cohort (Trap A)	Arm	clean	correlated	reversed	min(rev, corr)
Thyroid	JTT	0.629	0.829	0.417	0.417
Thyroid	mask	0.689	0.908	0.398	0.398
Thyroid	mask+JTT	0.699	0.888	0.455	0.455
Thyroid	U-MtE	0.729	0.824	0.622	0.622
Thyroid	U-MtE+JTT	0.629	0.848	0.363	0.363
Thyroid	balanced	0.717	0.757	0.677	0.677
Thyroid	mask+SPLINCE	0.603	0.599	0.610	0.599
Thyroid	U-MtE+bal.	0.798	0.804	0.787	0.787
Thyroid	U-MtE prot.	0.728	0.769	0.682	0.682
Thyroid	SPLINCE	0.577	0.565	0.587	0.565
Capsule	JTT	0.786	0.928	0.596	0.596
Capsule	mask	0.860	0.972	0.665	0.665
Capsule	mask+JTT	0.878	0.956	0.760	0.760
Capsule	U-MtE	0.815	0.972	0.539	0.539
Capsule	U-MtE+JTT	0.881	0.945	0.787	0.787
Capsule	balanced	0.847	0.916	0.760	0.760
Capsule	mask+SPLINCE	0.786	0.715	0.852	0.715
Capsule	U-MtE+bal.	0.899	0.939	0.839	0.839
Capsule	U-MtE prot.	0.855	0.970	0.659	0.659
Capsule	SPLINCE	0.741	0.668	0.788	0.668
Dermoscopy hair	JTT	0.745	0.898	0.564	0.564
Dermoscopy hair	mask	0.712	0.916	0.454	0.454
Dermoscopy hair	mask+JTT	0.711	0.880	0.509	0.509
Dermoscopy hair	U-MtE	0.735	0.913	0.506	0.506
Dermoscopy hair	U-MtE+JTT	0.729	0.885	0.541	0.541
Dermoscopy hair	balanced	0.785	0.850	0.722	0.722
Dermoscopy hair	mask+SPLINCE	0.598	0.572	0.621	0.572
Dermoscopy hair	U-MtE+bal.	0.778	0.860	0.695	0.695
Dermoscopy hair	U-MtE prot.	0.718	0.904	0.502	0.502
Dermoscopy hair	SPLINCE	0.600	0.575	0.637	0.575

Table 12: Remedy analyses regenerated with per-image predictions after the other runs (same commands and seeds; crossed bootstrap).

Analysis	Cohort	Contrast (Trap A, reversed)	Estimate [95% CI]
LaMa inpainting then masking	thyroid	mask_lama – mask	+0.224 [+0.206, +0.242]
	thyroid	mte_protect – mask_lama	+0.060 [+0.013, +0.105]
	thyroid	mte_balanced – mask_lama	+0.165 [+0.146, +0.185]
	ovary	mask_lama – mask	+0.128 [+0.084, +0.169]
	ovary	mte_protect – mask_lama	−0.085 [−0.120, −0.050]
U-MtE from real-debris templates	ovary	mte_balanced – mask_lama	+0.061 [+0.026, +0.095]
	capsule	mte – mask	−0.392 [−0.425, −0.360]
chest drains, RAD-DINO	capsule	mte_protect – mask	+0.003 [−0.014, +0.021]
	drains	mte – mask	+0.040 [+0.023, +0.057]
	drains	mte_protect – mask	+0.042 [+0.028, +0.058]
	drains	mte – jtt	+0.094 [+0.077, +0.111]
chest drains, DINOv2	drains	mte_balanced – balanced	−0.079 [−0.106, −0.044]
	drains	mte – mask	−0.001 [−0.009, +0.008]
	drains	mte_protect – mask	−0.003 [−0.013, +0.006]
	drains	mte – jtt	−0.005 [−0.021, +0.011]
	drains	mte_balanced – balanced	+0.031 [−0.023, +0.079]

Table 13: Remedy analyses whose per-image predictions were not saved, so their intervals could not be re-estimated with the crossed bootstrap: **point estimates only**.

Analysis	Cohort / run	Contrast (Trap A, reversed)	Point estimate
LaMa inpainting (oracle masks)	isic	lama – mask	+0.145
	isic	mask_lama – mask	+0.127
	isic	mte_protect – mask_lama	−0.086
	isic	mte_balanced – mask_lama	+0.123
text-prompted erasure	isic	lama – erm	+0.107
	thyroid	mask_text_erase – mask	+0.005
	thyroid	mte – mask	+0.204
	thyroid	mask_text_erase_balanced – balanced	+0.056
	ovary	mask_text_erase – mask	+0.008
	ovary	mte – mask	+0.215
	ovary	mask_text_erase_balanced – balanced	−0.010
	capsule	mask_text_erase – mask	+0.001
	capsule	mte – mask	−0.236
	capsule	mask_text_erase_balanced – balanced	+0.066
	hair (DermLIP)	mask_text_erase – mask	+0.004
	hair (DermLIP)	mte – mask	−0.076
SLAS	hair (DermLIP)	mask_text_erase_balanced – balanced	−0.112
	hair	tok_mask – tok_erm	+0.030
	hair	mts – tok_mask	+0.022
	hair	slas – tok_erm	+0.008
	hair	mts_oracle – tok_mask	+0.015
	hair	mts_balanced – tok_balanced	+0.051
	thyroid	tok_mask – tok_erm	+0.059
	thyroid	mts – tok_mask	+0.081
	thyroid	slas – tok_erm	+0.007
	thyroid	mts_oracle – tok_mask	+0.047
erasure + DFR (U10)	thyroid	mts_balanced – tok_balanced	+0.070
	thyroid	mte_dfr – dfr	+0.101
	thyroid	mte_dfr – mask_dfr	+0.007
	thyroid	mask_dfr – dfr	+0.095
	capsule	mte_dfr – dfr	+0.080
	capsule	mte_dfr – mask_dfr	+0.000
	capsule	mask_dfr – dfr	+0.080
	hair	mte_dfr – dfr	−0.039
	hair	mte_dfr – mask_dfr	+0.014
	hair	mask_dfr – dfr	−0.053
pseudo-group balancing (U8)	thyroid	umte_pbal reversed AUROC	0.663
	thyroid	mte reversed AUROC	0.622
	thyroid	pbal reversed AUROC	0.409
	thyroid	erm reversed AUROC	0.289
	hair	umte_pbal reversed AUROC	0.510
	hair	mte reversed AUROC	0.506
	hair	pbal reversed AUROC	0.498

Analysis	Cohort / run	Contrast (Trap A, reversed)	Point estimate
location-adaptive selection (U13)	hair	erm reversed AUROC	0.491
	thyroid	auto – mask	+0.395
	thyroid	auto – dfr	+0.057
	capsule	auto – mask	+0.168
	capsule	auto – dfr	+0.047
	hair	auto – mask	+0.251
	hair	auto – dfr	−0.038
	chest drains	auto – mask	+0.299
	chest drains	auto – dfr	−0.144
	thyroid	auto – mask	+0.398
split-validation selection (U14)	thyroid	auto – dfr	+0.060
	capsule	auto – mask	+0.187
	capsule	auto – dfr	+0.066
	ovary	auto – mask	+0.170
	ovary	auto – dfr	−0.007
	hair	auto – mask	+0.234
	hair	auto – dfr	−0.055
	chest drains	auto – mask	+0.417
	chest drains	auto – dfr	−0.026

5.9 Robustness of the remedy tests to leakage groups

Table 14. With the stricter embedding-based groups of Stage 4, the thyroid remedy tests are unchanged; in the ovary, erase > augment (P3) and U-MtE+balancing > balancing (P4) lose significance: “erase beats augment” is robust in thyroid and hair but fragile in the ovary.

Table 14: The remedy tests re-run with the stricter embedding-based leakage groups (cohorts without patient identifiers).

Cohort	Test	With embedding-based leakage groups	CI > 0
Thyroid	P2 U-MtE (protected) $\hat{\iota}$ mask	+0.251 [+0.215, +0.284]	yes
Thyroid	P3 erase $\hat{\iota}$ augment	+0.192 [+0.160, +0.223]	yes
Thyroid	P4 U-MtE+balanced $\hat{\iota}$ balanced	+0.152 [+0.127, +0.177]	yes
Ovary	P2 U-MtE (protected) $\hat{\iota}$ mask	+0.051 [+0.020, +0.079]	yes
Ovary	P3 erase $\hat{\iota}$ augment	+0.024 [−0.014, +0.062]	no
Ovary	P4 U-MtE+balanced $\hat{\iota}$ balanced	+0.038 [−0.005, +0.081]	no
Capsule	P2 U-MtE (protected) $\hat{\iota}$ mask	−0.006 [−0.022, +0.014]	no
Capsule	P3 erase $\hat{\iota}$ augment	−0.128 [−0.158, −0.098]	no
Capsule	P4 U-MtE+balanced $\hat{\iota}$ balanced	+0.101 [+0.071, +0.132]	yes

5.10 Fully fine-tuned networks

Table 15. Applying U-MtE *after* fine-tuning does nothing (−0.000 [−0.001, +0.001]): unconstrained fine-tuning builds a task-specific caliper feature that generic overlays do not move. Remedies must act *during* training. Each component alone works for only one architecture on thyroid (invariance for ResNet-50, +0.139 [+0.096, +0.184]; prediction consistency for ViT-S, +0.061 [+0.036, +0.084]). Combining projection, invariance and consistency (`umte_cons_ft`, registered before it was run) improves on fine-tuned masking for both: ResNet-50 +0.195 [+0.153, +0.237], ViT-S +0.119 [+0.036, +0.198] — without any artifact annotation. It fails on capsule (−0.026 [−0.056, +0.006]), the pathology-like regime, and is not significant on the ovary (+0.045 [−0.036, +0.130] with 15 clusters), where the fine-tuned network barely learns the task.

Table 15: Remedies for end-to-end fine-tuned networks: change in reversed-test AUROC versus fine-tuned masking (last column: versus the same fine-tuned body without erasure). Folds as bootstrap clusters; crossed CIs.

Fine-tuned network (Trap A)	invariance (mte.ft)	erase-while-fine-tuning (umte.ft)	prediction consistency (cons.ft)	all three (umte.cons.ft)	fine-tune, then erase
Thyroid, ResNet-50	+0.139 [+0.096, +0.184]	+0.186 [+0.144, +0.230]	+0.042 [+0.011, +0.074]	+0.195 [+0.153, +0.237]	−0.000 [−0.001, +0.001]
Thyroid, ViT-S/16	−0.000 [−0.026, +0.026]	+0.032 [−0.093, +0.144]	+0.061 [+0.036, +0.084]	+0.119 [+0.036, +0.198]	−
Capsule, ResNet-50	−0.027 [−0.051, −0.006]	−	−	−0.026 [−0.056, +0.006]	−
Ovary, ResNet-50 (15 clusters)	−	−	−	+0.045 [−0.036, +0.130]	−

5.11 Unaltered data

Table 16. On the natural test sets, of 20 remedy–test-set comparisons with masking (all pairs), 5 favour the remedy and 8 favour masking; the rest are not significant. Protected U-MtE is slightly *worse* than masking on the natural thyroid split (the registered claims N1 and N2 were not supported). The remedies repair the constructed shortcut of the traps; on data where the shortcut is weak, they mostly cost a little.

Table 16: Remedies on the natural test sets (AUROC, all pairs, crossed CIs): no remedy improves on masking uniformly.

Natural test set	U-MtE – mask	U-MtE protected – mask	balancing – mask	U-MtE + balancing – mask
Thyroid (official split)	+0.004 [−0.012, +0.020]	−0.021 [−0.034, −0.008]	−0.017 [−0.064, +0.030]	−0.005 [−0.023, +0.013]
ISIC, BCN held out	+0.007 [+0.004, +0.010]	−0.019 [−0.024, −0.014]	+0.022 [+0.012, +0.031]	+0.010 [+0.006, +0.015]
ISIC, HAM held out	+0.013 [+0.009, +0.017]	−0.006 [−0.016, +0.004]	−0.032 [−0.049, −0.013]	+0.018 [+0.009, +0.028]
ISIC, MSK held out	−0.008 [−0.014, −0.002]	−0.003 [−0.012, +0.006]	−0.018 [−0.041, +0.004]	−0.016 [−0.024, −0.008]
Capsule (held-out frames)	−0.005 [−0.011, +0.001]	−0.023 [−0.029, −0.017]	−0.086 [−0.109, −0.065]	−0.008 [−0.013, −0.001]

5.12 The decision guide

Table 17: What carries the shortcut decides the fix.

Regime	Example	Recommended	How to diagnose it
Artifact outside the ROI	calipers or rulers beside the lesion	ROI masking	measure the overlap r of detected artifacts
In-ROI, shortcut in the artifact’s pixels	calipers, rulers, synthetic tubes	mask, then U-MtE (no labels) or U-MtE with balancing (labels); LaMa if artifact masks exist	oracle removal of the artifact recovers most of the gap
In-ROI, artifact resembles the pathology	capsule debris versus fibrin	protected U-MtE (no labels); masking + DFR (labels)	large fitted artifact–disease overlap; unprotected erasure lowers clean AUROC
Shortcut carried by correlates	dermoscopic hair, chest drains	DFR or group balancing (labels); JTT without labels	oracle removal recovers little
Encoder ignores the out-of-ROI artifact	chest tube with RAD-DINO	masking unnecessary outside	reversed $\approx$ clean AUROC at no overlap
Fully fine-tuned network	ResNet-50, ViT-S	erase while fine-tuning (projection + invariance + consistency)	post hoc erasure is null

No row claims an improvement on unaltered data: the guide tells a practitioner which method fits which regime and how to diagnose the regime, not which method to use everywhere.

6 Validating the artifact labels: the clinician audit

Every trap depends on automatic artifact labels (a rule-based caliper detector, a debris probe, published semi-automatic hair masks) that have so far been checked only visually by a non-clinician. A blinded audit package is ready in `audit/`:

- **Sample.** 240 images: 4 cohorts $\times$ 3 cells (artifact inside the ROI, outside it, artifact-free, by the automatic labels) $\times$ 20 (10 positive, 10 negative diagnosis), fixed seed, the 518-px images the

models saw.

- **Blinding.** Every image and its overlay (ROI outline, automatic artifact mask) carry a neutral identifier; the key is kept outside the repository and fixed in advance by its SHA-256 (4b5db1521f1fc214...).
- **Task** (about 30–40s per image, 2.5h in total): artifact present?, location relative to the ROI?, is the red mask correct?, is the green ROI correct? The sheet is `audit/review.sheet.csv`.
- **Analysis.** `audit/analyse_audit.py` computes precision and recall of the automatic labels, location agreement, ROI correctness and, with a second reviewer, Cohen’s κ , and writes Table 18.
- **Licences** (Table 19). Images whose redistribution is not confirmed (TN3K thyroid images; overlays that draw the ISIC 2019 hair masks) are regenerated locally from the original data by the sampling script instead of being committed.

Rated so far: 0 of 240.

Table 18: Image audit of the automatic labels (to be filled).

Audit not yet performed. The table of precision, recall, location agreement, Cohen’s κ and ROI-mask correctness per cohort is written here by `audit/analyse_audit.py` once the review sheet has been filled.

Table 19: Licence decisions for publishing images (`audit/LICENCES.md`); unclear means not published.

Dataset	Licence (source)	Committed?	Attribution
ISIC 2018 Task 1/2, ISIC 2019 images (incl. HAM10000, BCN_20000, MSK)	CC BY-NC 4.0 (challenge.isic-archive.com/data)	Yes (raw images)	ISIC Challenge datasets 2018/2019; Tschandl et al. 2018 (HAM10000); Combalia et al. 2019 (BCN_20000); Codella et al. (MSK)
HAM10000 lesion segmentations	CC BY-NC 4.0 (Harvard Dataverse doi:10.7910/DVN/DBW86T)	Yes (as ROI outlines)	Tschandl et al. 2020
ISIC 2019 hair/ruler masks (Wegley et al., Scholars’ Mine doi:10.71674/man1-qa33)	no licence stated on the repository page	No — every overlay that draws these masks stays local	—
TN3K images, nodule masks; TNCD labels	not stated for the data; the GitHub repositories (TRFENet, ACL) license their *code* under MIT only	No — all thyroid images local	—
MMOTU OTU_2d	CC BY 4.0 (figshare doi:10.6084/m9.figshare.25058690)	Yes	Zhao et al. 2022, MMOTU
SEE-AI project dataset	CC BY 4.0 (Kaggle ‘capsule-olo/kyucapsule’, the project’s account)	Yes	Yokote et al. 2023/2024, SEE-AI project
Capsule expert contamination masks	CC BY 4.0 (figshare 27645021)	Yes	figshare 27645021
NIH ChestX-ray14	not used in the audit or example figures	—	—

7 The thesis manuscript

Title *Masking cannot remove what is inside the region of interest: location-dependent shortcut mitigation across dermoscopy, ultrasound and capsule endoscopy.* Target: *Medical Image Analysis*. Main text 26 pages (review format), supplement 28 pages; abstract 247 words (limit 250). An earlier version centred on U-MtE in NeurIPS format exists (`paper_neurips/`) but is not the current manuscript.

Structure Introduction; data and setup; the location law (controlled, real, transplant, matched, adjusted, replication including new patients, chest-radiograph boundary); mechanism; theory (held-out prediction); consequences on unaltered data (operating points; the thyroid subgroup labelled as exploratory confirmation); the decision guide; robustness; discussion with limitations. U-MtE, its ablations and the remedy results are in the supplement, with the registration table, the operating points of every cohort, the literature comparison, the covariate balance and the CLAIM checklist.

Why U-MtE is not the headline It repairs the constructed shortcut in the traps and fails to transfer uniformly to unaltered data (above). The finding that holds everywhere is the location law; the remedies are best presented as a decision guide.

Number audit Every interval in the main text and supplement must match an interval of a result file produced with the crossed bootstrap, together with its estimate; every other three-decimal number must occur in a result file. Result: 594 intervals and 1341 numbers checked, 0 failed to trace. The six progress reports are checked the same way.

CLAIM 2024 (Tejani et al., 2024) 43 items: 35 yes, 3 partial, 2 no, 3 not applicable (partial or no: items 13, 18, 21, 33, 36). The remaining gaps are acquisition protocols the providers did not release, the absence of external ultrasound and capsule cohorts, and the clinician audit.

Related work and novelty The closest prior work is the dermoscopy debiasing literature (Bisoto et al., 2020, 2023; Nauta et al., 2022), linear concept erasure (Belrose et al., 2023; Ravfogel et al., 2020), covariance-preserving erasure (Holstege et al., 2025) and biased-prompt projection (Chuang et al., 2023). What is new: the overlap as the controlled variable and the location law across modalities; the causal and theoretical account; generic overlays used to *estimate* a nuisance subspace rather than as augmentation; and the ordering mask-then-erase. Disease protection is an adaptation of covariance-preserving projection and is presented as such.

7.1 Draft submission statements (for the author to confirm)

Data availability All datasets are public: ISIC 2018, 2019 and 2020 (CC BY-NC 4.0), HAM10000 lesion masks (CC BY-NC 4.0), the published ISIC 2019 hair/ruler masks, TN3K with TNCD labels, MMOTU (CC BY 4.0), SEE-AI and its expert contamination masks (CC BY 4.0), NIH ChestX-ray14 and RANZCR-CLiP (competition terms). Derived masks, cohort files and all result files are in the repository; images are not redistributed where the licence does not allow it.

Code availability Code, registrations with their commit history, result files and the scripts that regenerate every table, figure and number are in the public repository.

Ethics Retrospective analysis of public, de-identified datasets released for research; no new data were collected and no patient was contacted. Whether a formal waiver is needed must be confirmed with the institution.

Use of AI assistance The experiments, analysis code, reports and parts of the manuscript text were produced with the assistance of an AI system (Claude, Anthropic) working under the author’s direction; the author designed the study, reviewed every result and is responsible for the content. The wording should follow the journal’s policy on generative-AI disclosure.

8 Limitations of the work as a whole

1. Artifact labels come from detectors and a probe, audited visually by a non-clinician; the clinician audit is pending.
2. Three cohorts publish no patient identifiers; near-duplicate and embedding grouping are proxies.
3. Reversed environments are constructed by subsampling; natural test sets carry weak shortcuts.
4. Frozen linear probes are below published fine-tuned models in absolute performance (thyroid).
5. External validation exists for dermoscopy only.
6. The theory is fitted per model and cannot see correlate-carried shortcuts.
7. Disease protection needs diagnosis labels of artifact-free images; balanced variants need image-level artifact labels.
8. Some secondary analyses (text prompts, SLAS, pseudo-groups, erasure + DFR, the selectors, LaMa on hair) could not be re-estimated with the crossed bootstrap and are reported as point estimates; the chest-radiograph devices and drains, LaMa on thyroid and ovary and the template variant were regenerated with per-image predictions.
9. Commit times of registrations are machine-set; only the registrations pushed to GitHub before running carry an external time stamp.

9 What failed or changed

- **No remedy transfers uniformly to unaltered data;** protected U-MtE did not meet its natural-data claims.
- **U-MtE fails for pathology-like artifacts** without protection, and for correlate-carried shortcuts at all.
- **Plain U-MtE failed on the dermoscopy ruler and on both MedSigLIP sweeps;** protection fixed only the debris one.
- **Post hoc erasure on fine-tuned networks is null;** single-component fine-tuning remedies transfer to one architecture only.
- **Text-prompted erasure is a near no-op.**
- **Erase > augment is fragile in the ovary** (not significant after Holm, and not with embedding groups).
- **The clinician audit has not been done.**
- **Two committed figures** (report/figures/data_thyroid.png, data_isic.png) predate the licence check and conflict with it; they need a decision.

10 Anticipated questions

Why erase instead of simply augmenting with the overlays? Augmentation teaches the head that the *generated* overlays are uninformative; the real artifact is a different pattern and keeps its association. Erasure removes the whole subspace in which foreign overlays move the representation, which also contains the real artifact’s direction when the two share it. The ablation (erase $\gg$ augment on thyroid with three encoders) confirms this.

Why generic overlays and not artifact templates? Templates need examples of the artifact, and an artifact detector to find them; the aim is a method that needs only what masking needs. Templates do help when available (the insert-then-erase precursor), but the generic library works for calipers in every encoder family without any artifact example.

Why does U-MtE fail on capsule, and why does protection fix it only partly? Debris and erosion fibrin share colour and texture, so the overlay subspace overlaps the disease (ρ large). Protection keeps the label directions estimated on artifact-free images, which removes the damage; it cannot create evidence that masking and erasure together removed, so it only reaches masking’s level.

Why does nothing pixel-based help for hair? Because most of the hair shortcut is not in the hair pixels (Stage 3: oracle removal of every hair patch gains little). A method that removes pixels or pixel-induced directions cannot remove an association carried by age, sex and body site.

Why is fine-tune-then-erase null but erase-while-fine-tuning not? After unconstrained fine-tuning on correlated data the network encodes the caliper along task-specific directions that generic overlays do not move; estimating the overlay subspace on the network’s own features *during* training, and requiring overlay-invariant features and predictions, stops that feature from forming.

If no remedy helps on unaltered data, what is the practical contribution? The practical contribution is the diagnosis: know where the artifacts sit, do not rely on masking when they sit inside the ROI, and pick the remedy by what carries the shortcut. On data with a strong in-ROI shortcut — the situation the traps construct and the thyroid operating points show — the right remedy makes a large difference.

Why keep SLAS and text prompts if they do little? Both were registered; negative results tell a reader which modern alternatives do not solve the problem, and SLAS’s oracle ceiling is itself evidence that hair is correlate-carried.

11 Conclusion

Supported. For frozen foundation-model features, U-MtE removes in-ROI shortcuts carried by distinct overlays without artifact annotation (thyroid with three encoder families and three sizes; ovary with MedSigLIP); erasure beats augmentation with the same overlays; protection makes it safe for pathology-like artifacts; with image-level labels, U-MtE with balancing or masking with DFR is the most robust choice. For fine-tuned networks, erasing while fine-tuning works on thyroid for both architectures. **Qualified.** No remedy improves on masking uniformly on unaltered data; the remedies are therefore a decision guide, not a headline method. **Not supported.** U-MtE for correlate-carried shortcuts; post hoc erasure after fine-tuning; text-prompted erasure. **Not yet available.** A clinician’s audit of the artifact labels; external validation in ultrasound and capsule endoscopy.

12 Questions for the supervisor

1. Should the remedies stay in the thesis as a decision guide within the main work, or form a separate chapter (or a second paper) with U-MtE as its contribution?
2. Who should perform the image audit (a dermatologist, a sonographer and a gastroenterologist?), and must it be completed before submission?

3. Is *Medical Image Analysis* the right target, or would a clinical-AI venue suit the operating-point results better?
4. Are the draft data-availability, ethics and AI-assistance statements acceptable to the institution?

13 Remaining work before submission

(i) Run the clinician audit and replace Table 18 with the output of `audit/analyse_audit.py`; (ii) decide on the two figures that conflict with the licence rule; (iii) confirm the submission statements; (iv) settle the supervisor’s decisions above; (v) deposit a snapshot of the repository with its registrations and tag the submitted version.

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